

# Tuscany Pyroconvection-Type Forecast – ICON-2I 2.2 km – VALID 2026-07-15

3-hourly, 24 h. Run 2026-07-15 00Z. Method after Castellnou et al. (2022), JGR-Atmos.

## How to read this product

Two panels are produced. The **fuel-gated** map is the expected, operationally comparable product (the equivalent of the Catalan “tipus de piroconveccio” map): a pyroCu/pyroCb class is assigned **only where a fire could actually reach  $\geq 10$  MW/m** of fireline intensity on the real Tuscany fuels. The **potential** map is an unconditional **upper bound** – it assumes a pyroCu-capable fire in *every* cell. Use the gated panel for situational awareness; use the potential panel only to see the atmospheric ceiling.

## Method and its limits (read this before using the classes)

The column is classified from a **real vertical profile**, not a standard atmosphere: per-cell heights from the model geopotential (FI), the mixing depth from the **bulk Richardson number** (first Rib  $\geq 0.33$  above 200 m AGL, referenced to the 2 m / 10 m state), the LCL from the exact **Bolton (1980)** formula, and the humidity at the ABL top from the model’s RH. The cap gamma-theta is taken over ABL+200 m to ABL+1200 m.

The **mixed-layer stability** diagnostic is the weakest link in this product, and is a *proxy*, not a measurement. Validated against IGRA radiosondes (JJA 12Z, period of record, n=2835), a 5-pressure-level column contains only 0-2 model levels inside the mixed layer – one or none in 92% of coastal columns – and the lowest sits in the superadiabatic surface layer. The mixed-layer dtheta/dz is therefore **not measurable from this archive**: fitting it across the in-mixed-layer levels has no skill (Youden J  $\sim 0.00$ ), and measuring theta up to the Rib ABL top folds in the entrainment jump (Delta-theta), which Castellnou et al. (2022, sec.2.1.1) treat as a variable *separate* from the gradient the ladder conditions on.

What is used instead is the **parcel mixing depth** as a proxy: at the  $1.1e-3$  K/m threshold the criterion reduces to “well-mixed layer  $\geq 455$  m deep”. It is the only candidate with skill (J = 0.29 inland / 0.55 coastal, r = +0.50 against the radiosonde truth). It **over-flags**: inland specificity is 0.29, so of the columns that are truly stable it still calls  $\sim 71\%$  pyroCu-capable (sensitivity 0.99 – it rarely misses a capable column).

**Consequence: read the classes as a screening flag, not as calibrated counts.** The class *totals* on these maps are not quantitatively trustworthy. The hybrid product (PYROCONV\_SOURCE=hybrid, ICON-EU model levels) measures this gradient properly; on this 5-level source, set PYROCONV\_ML\_METHOD=surface\_to\_abl or mid\_layer for the superseded measurements.

**Ladder actually used for this run: noshear.** The full method has a fifth diagnostic – the distance from the ABL/LCL to the height of maximum wind shear – which this product does not yet compute, so the four-diagnostic ladder runs. The shear clause is a *necessary* condition for the top class, so omitting it makes class 4 somewhat **easier** to reach here than in the full method. Treat class 4 as an alert to inspect the column, not as a calibrated probability.

The classes express atmospheric predisposition **given a fire of sufficient power**; in the gated panel that power is computed, not assumed. They are paper-informed thresholds, not locally validated ones.

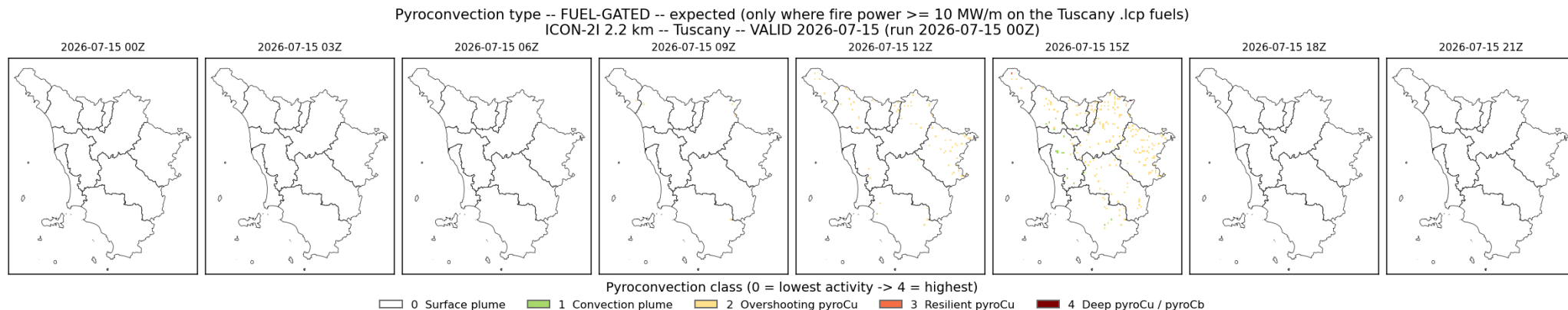


Figure 1: gated

## Expected pyroconvection type – FUEL-GATED

## Atmospheric potential – UPPER BOUND (assumes a pyroCu-capable fire in every cell)

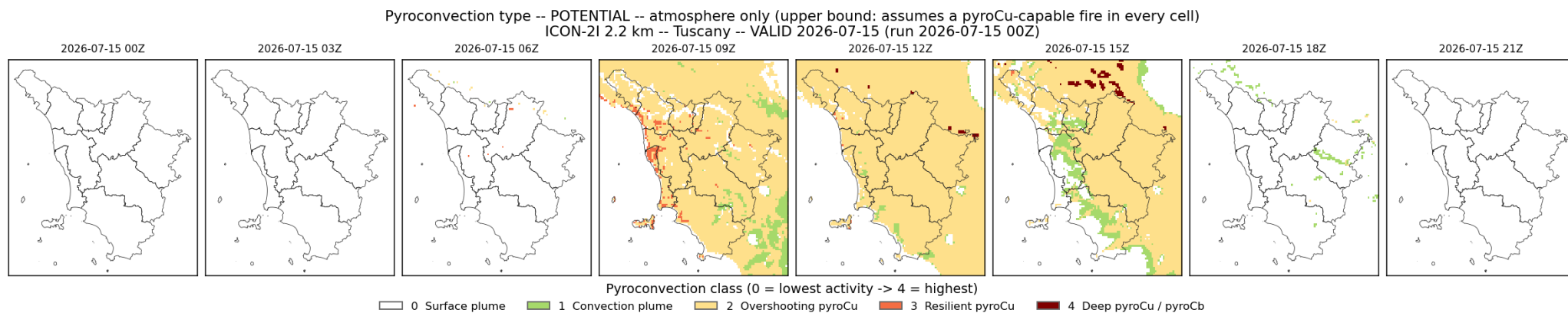


Figure 2: potential

## Class scale (low -> high pyroconvective activity)

| Level | Colour | Class         | Meaning                                                |
|-------|--------|---------------|--------------------------------------------------------|
| 0     | white  | Surface plume | Buoyant smoke plume; no significant cloud development. |

| Level | Colour   | Class                | Meaning                                                                       |
|-------|----------|----------------------|-------------------------------------------------------------------------------|
| 1     | green    | Convection plume     | Plume penetrates a stable mixed layer; condensation possible, no pyroCu.      |
| 2     | yellow   | Overshooting pyroCu  | Brief pyrocumulus; cloud base above the mixing height (LCL/ABL > 1).          |
| 3     | orange   | Resilient pyroCu     | Persistent pyrocumulus in an unstable column (LCL/ABL < 1).                   |
| 4     | dark red | Deep pyroCu / pyroCb | Deep pyroconvection / pyrocumulonimbus; weak upper cap lets the plume deepen. |

### Classification thresholds (ladder in use: **noshear**)

| Diagnostic                                                   | Threshold                                  | Effect                                                                                         |
|--------------------------------------------------------------|--------------------------------------------|------------------------------------------------------------------------------------------------|
| ML stability <i>proxy</i> : parcel mixing depth (see Method) | depth < 455 m (“stable”)                   | Convection plume only – no pyroCu                                                              |
| ML stability proxy                                           | depth >= 455 m (= dtheta/dz <= 1.1e-3 K/m) | Column is pyroCu-capable (weak filter: spec. 0.29 inland)                                      |
| LCL / ABL ratio                                              | 1.0 – 1.60                                 | Overshooting pyroCu (brief)                                                                    |
| LCL / ABL ratio                                              | < 1.0                                      | Resilient pyroCu (persistent)                                                                  |
| LCL / ABL ratio                                              | <= 1.10 (+ conditions below)               | Admissible for deep pyroCu / pyroCb                                                            |
| Cap gamma-theta (ABL+200 m -> ABL+1200 m)                    | <= 4.2e-3 K/m (weak cap)                   | Permits deepening to pyroCb                                                                    |
| Cap gamma-theta                                              | >= 4.8e-3 K/m (strong cap)                 | Inhibits deepening (resilient at most)                                                         |
| RH at the ABL top (mean, ABL +/- 150 m)                      | >= 80%                                     | Required for classes 3 and 4                                                                   |
| Shear-maximum distance / ABL                                 | <= 0.30                                    | Required for class 4 <b>in the 5-diagnostic ladder only</b> (not resolvable here – see Method) |
| Fireline intensity (fuel gate, gated panel)                  | >= 10 MW/m                                 | Minimum fire power for any pyroCu (Tedim et al. 2018)                                          |
| ABL depth                                                    | < 600 m                                    | Held at surface plume (mixing too shallow)                                                     |
| Usable pressure levels                                       | < 4                                        | Cell not classified                                                                            |

### Reference cases (Castellnou et al. 2022, Table 1)

The paper’s labelled events, which anchor the published 3-diagnostic ladder (`pyflam.pyroconvection_type(ladder="castellnou")`), still the library default and regression-tested against these rows). The profile ladders used for this map are stricter at the top: they additionally require a moist ABL top and an LCL close to it.

| Case  | Observed type        | LCL/ABL | ML dtheta/dz     | gamma-theta (700-500)  |
|-------|----------------------|---------|------------------|------------------------|
| T21   | Convection plume     | –       | stable           | –                      |
| SCQ32 | Overshooting pyroCu  | > 1     | neutral/unstable | –                      |
| M11   | Resilient pyroCu     | < 1     | unstable         | 4.2e-3 (resilient cap) |
| SCQ41 | pyroCu, not pyroCb   | < 1     | unstable         | 5.1e-3 (strong cap)    |
| SCQ51 | Deep pyroCu / pyroCb | < 1     | unstable         | 3.9e-3 (weak cap)      |

Forcing: ICON-2I 2.2 km full-Italy GRIB (MISTRAL / AgenziaItaliaMeteo, CC-BY), Tuscany subset: geopotential, temperature, RH and wind on 5 pressure levels plus the 2 m / 10 m state, surface pressure and orography. Classifier: pyflam.pyroconvection\_type (bulk-Richardson ABL, Bolton LCL, mixed-layer dtheta/dz proxy, cap gamma-theta, ABL-top RH; no surface CAPE). Gate: Rothermel + Cruz-2005 crown on the .lcp fuels with forecast moisture/wind. Per-hour diagnostic rasters (ABL, LCL, LCL/ABL, ML dtheta/dz, cap, RH-top) accompany the classes. Province borders: ISTAT-derived (openpolis geojson-italy). Generated by tests/pyroconv\_daily.py.